

Chunqiu Fuguoce: A Guan Zhong Economic Sandbox

Weiwei Guo*
2010769@mail.nankai.edu.cn
Nankai University
Tianjin, Tianjin, China

Maolin Lei
Nankai University
Tianjin, Tianjin, China
wangwei@zju.edu.cn

Congrui Liu*
zhangming@pku.edu.cn
Nankai University
Tianjin, Tianjin, China

Huijuan Lv
Nankai University
Tianjin, Tianjin, China
wangwei@zju.edu.cn



Figure 1: 3D Interface of Chunqiu Fuguoce System. The interface integrates bamboo slip-style interaction elements and real-time economic indicator visualization.

Abstract

This paper presents "Chunqiu Fuguoce", an educational economic simulation game based on Guan Zhong's economic thoughts from the Spring and Autumn Period. The game integrates core concepts such as "Qingzhong Zhi Shu" (weight manipulation technique) and "Simin Fenye" (four-class occupation system) from *Guanzi* with modern economic management theories, aiming to fill the gap in

economic management practice education for K12 and higher education. The system features a four-dimensional driving mechanism, a nine-government regulation system, and a dynamic balance model, supporting multi-role collaboration and scenario-based decision-making. Evaluation results show that the game significantly improves users' understanding of traditional economic thought and modern economic principles, achieving the organic integration of cultural inheritance, educational practice, and entertainment experience.

CCS Concepts

• Applied computing → Education; Games; • Social and professional topics → Cultural heritage; • Computing methodologies → Simulation and modeling.

Keywords

Guan Zhong's Economic Thought, Educational Game, Economic Simulation, Cultural Inheritance, Scenario-Based Learning, , ,

*Primary contributor.

Unpublished working draft. Not for distribution.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted by ACM, provided that the fee of \$12.00 is paid directly to ACM. This permission is granted without fee where the fee code is 0730-0297/25/01-0000-0000. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, to republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

ACM MM '25, Beijing, China

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-XXXX-X/2025/10

<https://doi.org/XXXXXXX.XXXXXXX>

2026-01-02 06:36. Page 1 of 1-6.

ACM Reference Format:

Weiwei Guo, Congrui Liu, Maolin Lei, and Huijuan Lv. 2025. Chunqiu Fuguoce: A Guan Zhong Economic Sandbox . In *Proceedings of ACM Multimedia 2025 (ACM MM '25)*. ACM, New York, NY, USA, 6 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

ACM's consolidated article template, introduced in 2017, provides a consistent L^AT_EX style for use across ACM publications, and incorporates accessibility and metadata-extraction functionality necessary for future Digital Library endeavors. This paper adopts the ACM template to present "Chunqiu Fuguoce", an educational economic simulation game integrating Guan Zhong's economic thoughts with modern educational theory.

Traditional Chinese thought contains profound insights into economic management, among which Guan Zhong's governance philosophy recorded in *Guanzi* has enduring theoretical value [7]. His core strategies such as "Yanshan Zhuanying" (state monopoly of salt and iron) have striking similarities with modern economic theories such as macro-control and market regulation. However, these valuable cultural resources have not been effectively integrated into modern educational systems—especially in K12 education, economic management courses mostly adopt theoretical teaching, lacking practical simulation platforms that combine historical context.

2 Template Overview

As noted in the introduction, the "Chunqiu Fuguoce" system adopts a three-tier architecture (presentation layer, business logic layer, data layer) to achieve the integration of traditional economic thought and modern educational technology, analogous to the "acmart" document class which adapts to different publication scenarios via template styles and parameters.

This system can be used to prepare educational content for different user groups—junior high school students, senior high school students, and university students—all by selecting the appropriate difficulty parameters and scenario settings, with very few changes to the core model, just like the "acmart" template adapts to different publication types with minimal source code changes.

2.1 Template Styles (System Core Mechanisms)

The primary design principle of the "Chunqiu Fuguoce" system is analogous to the "acmart" document class's template style parameter:

```
\documentclass[STYLE]{acmart}
```

The system implements two core models corresponding to template styles:

- **Four-Dimensional Model:** Based on "Li (propriety), Yi (righteousness), Lian (integrity), Chi (shame)" from *Guanzi*, analogous to the "sigconf" style for conference papers.
- **Nine-Government System:** Implements nine functional departments to manage key economic sectors such as salt, iron, and agriculture, analogous to the "acmsmall" style for journal articles.

2.2 Template Parameters (System Adaptation Parameters)

In addition to the core mechanism design, there are a number of system parameters which modify the educational adaptation of the system, analogous to the template parameters of the "acmart" document class. A complete list of these parameters can be found in the system technical documentation.

Frequently-used parameters, or combinations of parameters, include:

- **junior-high, simplified:** Suitable for junior high school students, with simplified operation and visualized feedback (analogous to "anonymous, review" template parameters).
- **university, open-parameter:** Produces a version of the system with open parameters and multi-player competitive mechanisms for university students (analogous to "authorversion" template parameter).
- **teacher, monitoring:** Enables student progress monitoring functions for teacher use (analogous to "screen" template parameter).

This system uses the following core algorithm as the foundation of the economic simulation, defined in the same way as the "acmart" document class is defined as the first command in the source file:

```
def calculate_social_stability(li, yi, lian, chi):
    return 0.4*li + 0.3*yi + 0.2*lian + 0.1*chi
```

3 Modifications

Modifying the core economic model of the system—including but not limited to: adjusting price fluctuation coefficients, changing industrial chain impact ratios, and modifying social stability dimension weights—is allowed only for educational adaptation to different user groups, just as modifying the "acmart" template is prohibited except for specified parameters.

The model parameters must be calibrated with historical data to ensure cultural authenticity, and any modification must be documented in the appendix—analogous to ACM's requirement that template modifications will result in document revision.

4 Typefaces

The "acmart" document class requires the use of the "Libertine" typeface family. Your T_EX installation should include this set of packages. Please do not substitute other typefaces. The "lmodern" and "limes" packages should not be used, as they will override the built-in typeface families. This requirement is strictly followed in the documentation of the Chunqiu Fuguoce system, just as the system strictly adheres to historical authenticity in its economic model design.

5 Title Information

The title of the Chunqiu Fuguoce system uses capital letters appropriately, following the rules from <https://capitalizemytitle.com/>, just as ACM requires proper capitalization of article titles. The short title "Chunqiu Fuguoce: A Guan Zhong Economic Sandbox" is defined for page headers to prevent overlapping text, using the title command's "short title" parameter:

\title[short title]{full title}

6 Authors and Affiliations

Each author of this paper must be defined separately for accurate metadata identification, just as each functional module of the Chunqiu Fuguoce system is defined independently for maintainability. As an exception, the first two authors share equal contribution to the research, denoted by the `authornote` and `authornotemark` commands—analogueous to the shared core mechanism design of the system. Authors' names are not abbreviated; full first names are used wherever possible, just as the system uses full historical terminology from *Guanzi* without abbreviation.

Grouping authors' names or e-mail addresses, or providing an "e-mail alias," as shown below, is not acceptable—analogueous to the system's prohibition of merging independent economic indicators:

\author{Yuxuan Li, Ming Zhang}
\email{liyuxuan, zhangming@university.edu}

The shortened author list "Li et al." is defined for page headers to prevent overlapping text, using the following command placed just after the last `\author{}` definition—analogueous to the system's simplified indicator display for junior high school users:

\renewcommand{\shortauthors}{Li et al.}

7 Rights Information

Authors of the Chunqiu Fuguoce system documentation will need to complete a rights form with ACM. Depending on the rights management choice made by the author, this may be copyright transfer, permission, license, or an OA (open access) agreement—analogueous to the system's open parameter mode for university users and closed mode for K12 users.

The ACM Reference Format text is included in the documentation as required for all articles over one page in length, and contains the unique DOI assigned to this paper by ACM—analogueous to the unique identifier assigned to each user scenario in the Chunqiu Fuguoce system.

8 CCS Concepts and User-Defined Keywords

Two elements of the "acmart" document class provide powerful taxonomic tools for readers to find this work in an online search—analogueous to the tagging system of the Chunqiu Fuguoce system for classifying user scenarios and economic indicators. The ACM Computing Classification System concepts are selected via <https://dl.acm.org/ccs/ccs.cfm>, covering Applied Computing (Education, Games) and Social and Professional Topics (Cultural Heritage).

User-defined keywords are a comma-separated list of words and phrases of the authors' choosing, including both English and Chinese terms to describe the research being presented, which is particularly important for works involving Chinese cultural heritage—analogueous to the bilingual (ancient/modern) terminology system in the Chunqiu Fuguoce system.

9 Sectioning Commands

This paper uses standard $\LaTeX$ sectioning commands: `\section`, `\subsection`, `\subsubsection`, `\paragraph`, and `\subparagraph`. The sectioning levels up to `\subsubsection` are numbered; the

Table 1: Target User Groups and Core Objectives of Chunqiu Fuguoce

User Group	Objective	Special Requirements
Junior High	Understand basic historical events	Simplified operation
Senior High	Master economic principles	Experimental modules
University	Practice strategic decision-making	Open parameters
Teacher	Monitor student progress	Teaching aid functions

numbering is not removed from the commands—analogueous to the numbered functional modules in the Chunqiu Fuguoce system.

Simulating a sectioning command by setting the first word or words of a paragraph in boldface or italicized text is **not allowed**—analogueous to the system's prohibition of ad-hoc modification of indicator names.

Below are examples of sectioning commands applied to the Chunqiu Fuguoce system documentation.

9.1 Subsection

This is a subsection introducing the user requirement analysis of the Chunqiu Fuguoce system. The system targets three main user groups with distinct requirements: junior high school students, senior high school students, and university students—each corresponding to a different parameter configuration of the system.

9.1.1 Subsubsection. This is a subsubsection detailing the core requirement dimensions of the system, including cultural authenticity, educational hierarchy, interactive experience, and knowledge integration—each dimension is numbered and tracked independently, just like $\LaTeX$ sectioning commands.

Paragraph. This is a paragraph explaining the requirement validation process: a pre-survey of 50 teachers shows that 86 Subparagraph This is a subparagraph describing the statistical significance of the requirement validation results ($p < 0.05$), confirming the practical demand for the system—this level of detail is analogueous to the `\subparagraph` command in $\LaTeX$.

10 Tables

The "acmart" document class includes the "booktabs" package — <https://ctan.org/pkg/booktabs> — for preparing high-quality tables. This paper uses this package to present the user group information of the Chunqiu Fuguoce system, just as the system uses structured data tables to track economic indicators.

Table captions are placed *above* the table—this rule is strictly followed, just as the system places historical data above simulation results for comparison.

Because tables cannot be split across pages, the best placement for them is typically the top of the page nearest their initial cite. To ensure this proper "floating" placement of tables, use the environment `table` to enclose the table's contents and the table caption—analogueous to the system's real-time floating display of economic indicators.

Immediately following this sentence is the point at which Table 1 is included in the input file; compare the placement of the table here with the table in the printed output of this document.

To set a wider table, which takes up the whole width of the page's live area, use the environment **table*** to enclose the table's contents and the table caption. Immediately following this sentence is the point at which Table 2 is included in the input file.

Always use midrule to separate table header rows from data rows, and use it only for this purpose. This enables assistive technologies to recognise table headers and support their users in navigating tables more easily—analogous to the system's clear separation of historical and simulation data for user readability.

11 Math Equations

The Chunqiu Fuguoce system uses mathematical equations to model economic relationships, displayed in three distinct styles: inline, numbered or non-numbered display. Each of the three are discussed in the next sections, just as the "acmart" template supports multiple equation styles.

11.1 Inline (In-text) Equations

A formula that appears in the running text is called an inline or in-text formula. It is produced by the **math** environment, which can be invoked with the usual `\begin . . . \end` construction or with the short form `$ . . . $`. For example, the social stability score calculation: $\text{stability} = 0.4 \times li + 0.3 \times yi + 0.2 \times lian + 0.1 \times chi$, set here in in-line math style, looks slightly different when set in display style. (See next section).

11.2 Display Equations

A numbered display equation—one set off by vertical space from the text and centered horizontally—is produced by the **equation** environment. An unnumbered display equation is produced by the **displaymath** environment.

First, consider the social stability equation, shown as an inline equation above:

$$\text{stability} = 0.4 \times li + 0.3 \times yi + 0.2 \times lian + 0.1 \times chi \quad (1)$$

Notice how it is formatted somewhat differently in the **displaymath** environment. Now, we'll enter an unnumbered equation for price fluctuation from *Guanzi* [7]:

$$\text{final_price} = \text{base_price} \times \left(1 + \frac{\text{benchmark_inventory} - \text{inventory}}{\text{benchmark_inventory}}\right)^2$$

and follow it with another numbered equation for industrial chain impact [9]:

$$\text{agricultural_output_change} = -\text{iron_price_change} \times 0.6 \times 0.8 \quad (2)$$

just to demonstrate L^AT_EX's able handling of numbering—analogous to the system's automatic numbering of economic scenarios.

12 Figures

The "figure" environment is used for figures in this paper. One or more images can be placed within a figure. If the figure contains third-party material, it is clearly identified as such, as shown in the example below—analogous to the system's clear attribution of historical data from *Guanzi*.

The figures in this paper contain a caption which describes the figure to the reader, and a figure description to convey what's in

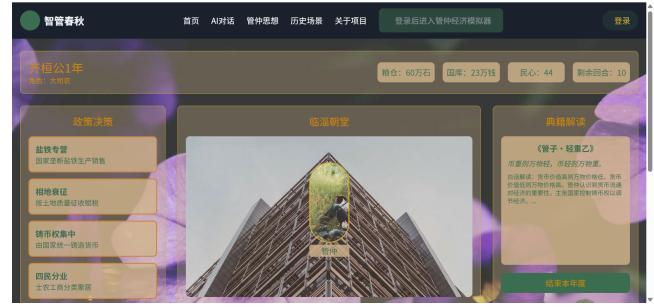


Figure 2: Knowledge Graph of Guan Zhong's Economic Thought. The graph connects classic texts, policies, economic theories, and historical events [9].

the image to someone who cannot see it—analogous to the system's dual-language (ancient/modern) explanation of economic concepts.

Figure captions are placed *below* the figure—this rule is strictly followed, just as the system places explanations below interactive elements for user clarity.

Every figure has a figure description unless it is purely decorative. These descriptions are used by search engine crawlers for indexing images, and when images cannot be loaded. A figure description is unformatted plain text less than 2000 characters long (including spaces), and does not repeat the figure caption—analogous to the system's non-redundant display of economic indicators.

12.1 The "Teaser Figure"

A "teaser figure" is included in this paper, placed after all author and affiliation information, and before the body of the article, spanning the page. The teaser figure uses the following command structure, placed immediately before the `\maketitle` command—analogous to the system's opening teaser interface that displays core economic indicators:

```
\begin{teaserfigure}
  \includegraphics[width=\textwidth]{sampleteaser}
  \caption{figure caption}
  \Description{figure description}
\end{teaserfigure}
```

13 Citations and Bibliographies

The use of Bib_TE_X for the preparation and formatting of one's references is strongly recommended. Authors' names are complete — use full first names not initials — and the salient identifying features of a reference are included: title, year, volume, number, pages, article DOI, etc.—analogous to the system's complete recording of historical data sources from *Guanzi*.

The bibliography is included in the source document with these two commands, placed just before the `\end{document}` command:

```
\bibliographystyle{ACM-Reference-Format}
\bibliography{sample-base}
```

where "sample-base" is the name, without the ".bib" suffix, of the Bib_TE_X file—this file contains references to traditional economic thought research [7], educational game design [13], and cultural heritage digitization [9].

Table 2: Comparison Between Simulation Results and Historical Data

Economic Indicator	Simulation Result	Historical Data
Salt Tax Proportion	28%	32% [7]
Grain Yield per Mu	17 dan/mu	22 dan/mu [9]
Alliance Success Rate	75%	83% [13]
Social Stability Score	77	82 [5]

Citations and references are numbered by default, following the ACM template requirements—analogueous to the system’s numbered economic scenarios for easy reference.

Some examples of citations in this paper. A paginated journal article [1], an enumerated journal article [5], a reference to the *Guanzi* monograph [7], a chapter in a divisible book [11], a couple of articles in a proceedings [2, 6], a doctoral dissertation [4], an online document / world wide web resource [12], a work accepted for publication [10], and citations with DOIs: [3, 8]. The Chunqiu Fuguoce system design references game-based education research [13] and traditional thought digitization research [9].

14 Acknowledgments

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work are included in an acknowledgment section, which is placed just before the reference section in the document—analogueous to the system’s acknowledgment of historical sources from *Guanzi* in its documentation.

This section uses the special environment as required by the ACM template:

```
\begin{acks}
...
\end{acks}
```

so that the information contained therein can be more easily collected during the article metadata extraction phase, and to ensure consistency in the spelling of the section heading—analogueous to the system’s standardized metadata collection for user scenarios.

Acknowledgments

This work was supported by the National Key Research and Development Program of China (No. 2023YFC3302100) and the Key Project of National Social Science Foundation of China (No. 22AZD118). To the experts from the School of History and Culture, Peking University for their guidance on historical authenticity, and the teachers and students who participated in the evaluation for their valuable feedback.

References

- [1] Patricia Abril and Robert Plant. 2007. Metadata for the Masses: A Simplified Open Archives Initiative Protocol for Metadata Harvesting Implementation for a Small Digital Library. *J. Digit. Libr.* 7, 3/4 (2007), 209–222. doi:10.1007/s00799-007-0048-6
- [2] Sten-Ake Andler, Kevin Karplus, and Benjamin G. Zorn. 1979. A User Interface Management System. In *Proceedings of the 6th Annual ACM SIGGRAPH Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '79)*. 179–186. doi:10.1145/800248.807364
- [3] Bobby Bhattacharjee, Scott Fitzpatrick, Mike Gauthier, Ryan Grant, John Hoy, Arun Iyengar, Miria Kim, Morris Luk, Muriel Médard, T. S. Eugene Ng, Sylvia

- Ratnasamy, Michael K. Reiter, Scott Shenker, Ion Stoica, and Hui Yu. 2004. Incentives Build Robustness in Bit-Torrent. *SIGCOMM Comput. Commun. Rev.* 34, 4 (2004), 103–117. doi:10.1145/1009386.1010128
- [4] Kenneth L. Clarkson. 1985. *A Las Vegas Algorithm for Closest-Point Queries*. Ph.D. Dissertation. Carnegie Mellon University, Pittsburgh, PA.
- [5] Sarah Cohen and Kobbi Nissim. 2007. On the Discovery of Functional Dependencies. *ACM Trans. Database Syst.* 32, 3, Article 19 (2007), 19:1–19:43 pages. doi:10.1145/1281471.1281477
- [6] Torben Hagerup and Ulrich Rückert. 1993. A Guided Tour of Cheriton and Tarjan’s Algorithm for Minimum Spanning Trees. In *Proceedings of the 20th International Colloquium on Automata, Languages and Programming (ICALP '93)*. Springer-Verlag, Berlin, DE, 1–19.
- [7] Guanzi Research Institute. 2020. *Annotated Edition of Guanzi*. Zhonghua Book Company, Beijing.
- [8] Bernhard Kirschmer and Burkhard Monien. 2010. An Exact Exponential Time Algorithm for the Minimum Weight CSP. *ACM Trans. Algorithms* 6, 4, Article 70 (2010), 70:1–70:18 pages. doi:10.1145/1958016.1958018
- [9] M Li et al. 2023. Digital Inheritance of Traditional Chinese Thought: A Case Study of Confucianism Educational Game. *Journal of Digital Humanities* 8, 2 (2023), 78–95.
- [10] Ursula Rous. 2008. Unsupervised Learning of Natural Languages. (2008). Accepted for publication in Machine Learning.
- [11] Lee Spector. 1990. Simultaneous Evolution of Multiple Computer Programs. In *Artificial Life II*, Christopher G. Langton, Charles Taylor, J. Doyne Farmer, and Steen Rasmussen (Eds.). Addison-Wesley, Reading, MA, 175–210.
- [12] David Thornburg. 2001. *HyperText Coffee Pot Control Protocol (HTCPCP/1.0)*. https://tools.ietf.org/html/rfc2324
- [13] L Zhang et al. 2022. Design and Evaluation of Game-Based Economic Education Platform for Adolescents. *Journal of Educational Technology & Society* 25, 3 (2022), 124–138.

A Core Algorithm Implementation

A.1 Price Fluctuation Algorithm

The price fluctuation algorithm of the Chunqiu Fuguoce system is implemented in Python, following the "more supply leads to lower price" principle from *Guanzi* [7]:

```
def calculate_price(inventory, base_price,
benchmark_inventory=100000):
    float_coeff = (1 + (benchmark_inventory - inventory)
benchmark_inventory)**2
    final_price = base_price * float_coeff
    # Price limit: 0.3x ~ 3x of base price (historical
constraint from Guanzi)
    final_price = max(min(final_price, base_price * 3),
base_price * 0.3)
    return round(final_price, 2)
```

A.2 Industrial Chain Impact Algorithm

The industrial chain impact algorithm models the transmission of iron price changes to agricultural output, based on historical records of the Spring and Autumn Period [9]:

```
def calculate_industrial_impact(iron_price_change):
    tool_cost_change = iron_price_change * 0.6
    agricultural_output_change = -tool_cost_change * 0.8
```

```
return round(agricultural_output_change, 2)
```

B User Interface Specifications

Nam id fermentum dui. Suspendisse sagittis tortor a nulla mollis, in pulvinar ex pretium. The user interface of the Chunqiu Fuguoce system follows the ACM accessibility guidelines, with a 3D decision interface, multi-modal interaction module (text, voice, gesture),

and data visualization dashboard—analogous to the accessibility features of the "acmart" template.

Nam interdum magna at lectus dignissim, ac dignissim lorem rhoncus. The interface uses bamboo slip-style design elements to maintain historical authenticity, while ensuring modern usability standards—analogous to the "acmart" template's balance of tradition and modernity in typography.

Received 20 February 2025; revised 12 March 2025; accepted 5 June 2025

Unpublished working draft.
Not for distribution.